

ELIAS MARCALWA

Advanced SQL Null Functions

Q1

```
SELECT employee_id,  
       name,  
       COALESCE(Salary, 0) AS salary_with_default  
FROM employees;
```

employee_id	name	Salary_with_default
101	John	5000
102	Sarah	0
103	Mike	7500

Q2

```
SELECT employee_id,  
       name,  
       IFNULL(department, 'Not Assigned') AS department_name  
FROM employees;
```

employee_id	name	department_name
101	John	IT
102	Sarah	Not Assigned
103	Mike	Finance

Q3 Select order_id,
Customer_id
From orders
WHERE ISNULL(customer_id);

Output

order_id	Customer_id
1002	NULL
1005	NULL

Q4 SELECT order_id,
Customer_id
IFNULL(CAST(delivery_date AS String), 'pending') AS delivery_status
From orders;

Order_id	Customer_id	delivery_status
1001	201	2026-07-20
1002	202	pending
1003	203	2026-07-22

Q5 Select student_id,
name,
IFNULL(Final_grade, 0) AS final_grade
From students;

Student_id	name	final_grade
1	Alice	85
2	Bob	0
3	Carol	92

Q6

SELECT COUNT(*) AS not_graded_Count
From students
WHERE final_grade ~~0~~ ISNULL;

Output

not_graded_Count
2

Q7 Select productid,
name
Price - COALESCE (discount, 0) AS final price
From products;

Productid	name	final price
101	Laptop	950
102	Mouse	25
103	Keyboard	45

Q8 SELECT COUNT(*) AS Missing_email_count
FROM Customers
WHERE email IS NULL;

Output

Missing_email_count
3

Q9 Select customerid,
name,
COALESCE (email, 'No Email') AS email_display
From Customers;

Q10 SELECT payment_id,
COALESCE (method, 'unknown') AS method_display,
Status
FROM payments;

Payment_id	Method_display	Status
1	Credit Card	Completed
2	unknown	pending
3	Cash	Completed
4	Debit Card	failed
5	Unknown	Completed

Q11 SELECT Item_id,
Item_name,
NVL (quantity, 0) AS quantity_checked
FROM items;

Item_id	Item_name	Quantity_checked
101	Pen	50
102	Notebook	0
103	Eraser	25
104	Pencil	0
105	Marker	12

Q12 SELECT emp_id,
COALESCE (bonus, Commission) AS first_available_reward
FROM employees;

Output ->

emp_id	first_available_reward
101	500
102	300
103	700
104	Null
105	450

Q13

SELECT Count (*) AS no_room_Count
FROM classes
WHERE room IS NULL;

No room Count

3

Q14

SELECT student_id,
date,
COALESCE (Status, 'Not marked') AS attendance_Status
FROM attendance;

Student id	Date	attendance_Status
101	2026-07-20	Present
102	2026-07-20	Absent
103	2026-07-20	Not Marked
104	2026-07-21	Present
105	2026-07-21	Not Marked

Q15

SELECT account_id,
COALESCE (account_type, 'unknown') AS type_display,
COALESCE (balance, 0) AS balance_Checked
FROM accounts;

Output of Q15

account_id	type_display	balance_checked
1001	Savings	1500
1002	Unknown	2500
1003	Checking	0
1004	Unknown	0
1005	Business	5000

Q16 Select project_id,
title,
NVL(CAST(start_date AS STRING), 'TBD') AS start_display
From projects;

Project_id	title	Start_display
101	Website Redesign	2026-08-01
102	Mobile App	TBD
103	Data Migration	2026-09-15
104	CRM Implementation	TBD
105	Security Upgrade	2026-10-01

Q17 Select review_id,
product_id,
COALESCE(comment, 'No comment') AS comment_display,
COALESCE(rating, 0) AS rating_display
From reviews;

Output	review_id	product_id	comment_display	rating_display
1		101	Great Product	5
2		102	No comment	4
3		103	Good Value	0
4		104	No comment	0
5		105	Fast delivery	5

Q18 Select Supplier_id,
name,
COALESCE (phone, alt_phone, 'No Contact') AS Contact_number
From Suppliers;

Output

Supplier_id	name	Contact_number
101	ABC Supphes	SSS_1234
102	Global trading	SSS_9876
103	Fresh Foods LTD	No Contact
104	TechSource	SSS_4567
105	Office Depot	SSS_7890

Q19 Select user_id,
COALESCE (theme, 'light') AS theme_Set,
COALESCE (language, 'English') AS language_Set,
COALESCE (timezone, 'UTC') AS timezone_Set
From users;

Output

user_id	theme_Set	language_Set	timezone_Set
101	Dark	English	UTC+2
102	light	French	UTC
103	light	English	UTC
104	Dark	English	EST
105	light	Spanish	UTC

Q20

SELECT record_id,

machine_id,

COALESCE(issue, 'unknown issue') AS issue_log,

COALESCE(technician, 'Not Assigned') AS technician_name

FROM Maintenance_log;

output

record_id	machine_id	issue_log	technician_name
1	M101	oil leak	John Smith
2	M102	unknown issue	Sarah Jones
3	M103	Motor failure	Not assigned
4	M104	unknown issue	Not assigned
5	M105	Belt Replacement	Michael Brown